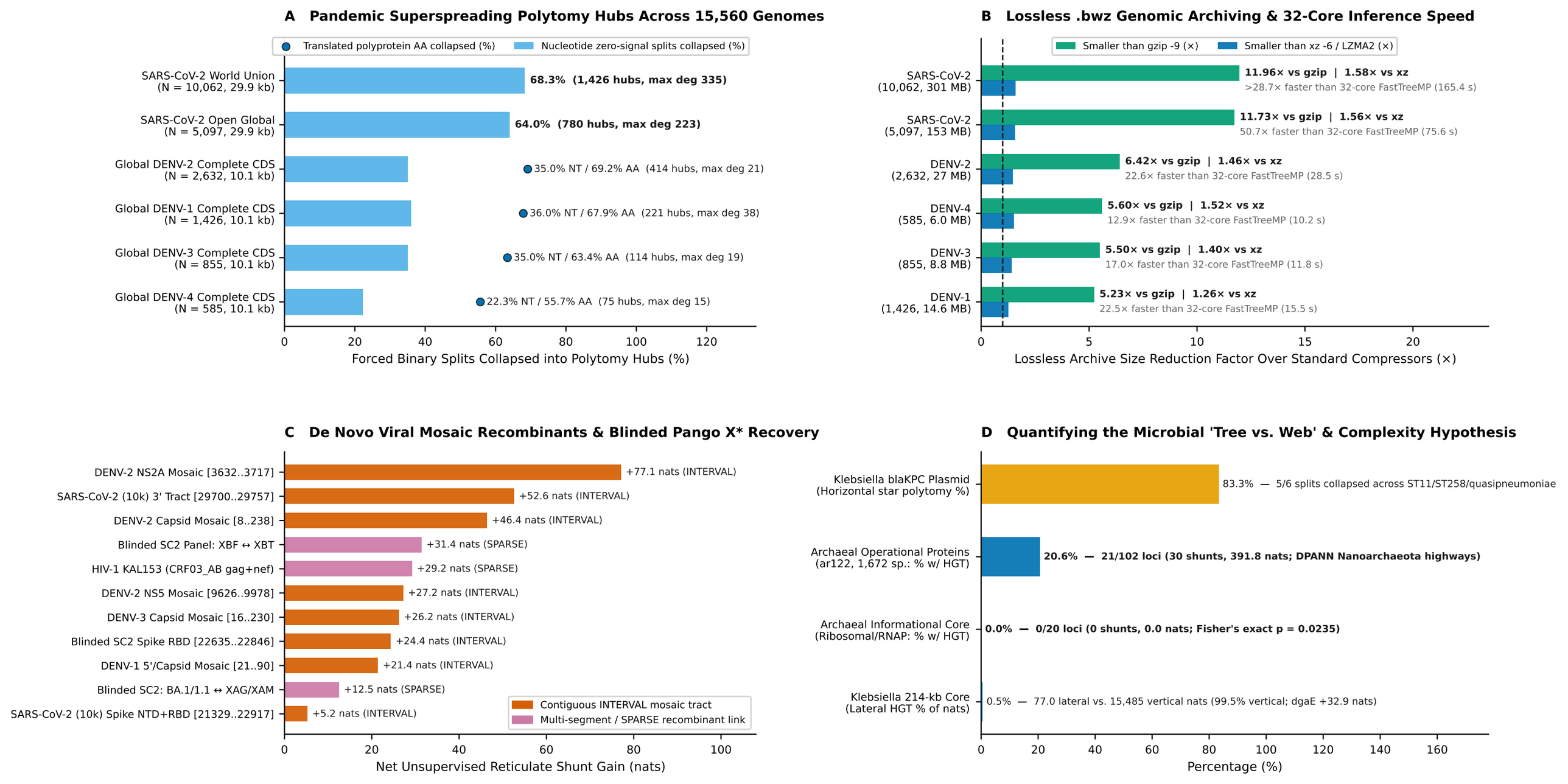
A Pandemic Superspreading Polytomy Hubs Across 15,560 Genomes
Translated polyprotein AA collapsed (%)
Nucleotide zero-signal splits collapsed (%)
SARS-CoV-2 World Union (N = 10,062, 29.9 kb)
68.3% (1,426 hubs, max deg 335)
SARS-CoV-2 Open Global (N = 5,097, 29.9 kb)
64.0% (780 hubs, max deg 223)
Global DENV-2 Complete CDS (N = 2,632, 10.1 kb)
35.0% NT / 69.2% AA (414 hubs, max deg 21)
Global DENV-1 Complete CDS (N = 1,426, 10.1 kb)
36.0% NT / 67.9% AA (221 hubs, max deg 38)
Global DENV-3 Complete CDS (N = 855, 10.1 kb)
35.0% NT / 63.4% AA (114 hubs, max deg 19)
Global DENV-4 Complete CDS (N = 585, 10.1 kb)
22.3% NT / 55.7% AA (75 hubs, max deg 15)
0
20
40
60
80
100
120
Forced Binary Splits Collapsed into Polytomy Hubs (%)
B Lossless .bwz Genomic Archiving & 32-Core Inference Speed
Smaller than gzip -9 (×)
Smaller than xz -6 / LZMA2 (×)
SARS-CoV-2 (10,062, 301 MB)
11.96× vs gzip | 1.58× vs xz
>28.7× faster than 32-core FastTreeMP (165.4 s)
SARS-CoV-2 (5,097, 153 MB)
11.73× vs gzip | 1.56× vs xz
50.7× faster than 32-core FastTreeMP (75.6 s)
DENV-2 (2,632, 27 MB)
6.42× vs gzip | 1.46× vs xz
22.6× faster than 32-core FastTreeMP (28.5 s)
DENV-4 (585, 6.0 MB)
5.60× vs gzip | 1.52× vs xz
12.9× faster than 32-core FastTreeMP (10.2 s)
DENV-3 (855, 8.8 MB)
5.50× vs gzip | 1.40× vs xz
17.0× faster than 32-core FastTreeMP (11.8 s)
DENV-1 (1,426, 14.6 MB)
5.23× vs gzip | 1.26× vs xz
22.5× faster than 32-core FastTreeMP (15.5 s)
0
5
10
15
20
Lossless Archive Size Reduction Factor Over Standard Compressors (×)
C De Novo Viral Mosaic Recombinants & Blinded Pango X* Recovery
DENV-2 NS2A Mosaic [3632..3717]
+77.1 nats (INTERVAL)
SARS-CoV-2 (10k) 3' Tract [29700..29757]
+52.6 nats (INTERVAL)
DENV-2 Capsid Mosaic [8..238]
+46.4 nats (INTERVAL)
Blinded SC2 Panel: XBF ↔ XBT
+31.4 nats (SPARSE)
HIV-1 KAL153 (CRF03_AB gag+nef)
+29.2 nats (SPARSE)
DENV-2 NS5 Mosaic [9626..9978]
+27.2 nats (INTERVAL)
DENV-3 Capsid Mosaic [16..230]
+26.2 nats (INTERVAL)
Blinded SC2 Spike RBD [22635..22846]
+24.4 nats (INTERVAL)
DENV-1 5'/Capsid Mosaic [21..90]
+21.4 nats (INTERVAL)
Blinded SC2: BA.1/1.1 ↔ XAG/XAM
+12.5 nats (SPARSE)
SARS-CoV-2 (10k) Spike NTD+RBD [21329..22917]
+5.2 nats (INTERVAL)
Contiguous INTERVAL mosaic tract
Multi-segment / SPARSE recombinant link
0
20
40
60
80
100
Net Unsupervised Reticulate Shunt Gain (nats)
D Quantifying the Microbial 'Tree vs. Web' & Complexity Hypothesis
Klebsiella blaKPC Plasmid (Horizontal star polytomy %)
83.3% — 5/6 splits collapsed across ST11/ST258/quasipneumoniae
Archaeal Operational Proteins (ar122, 1,672 sp.: % w/ HGT)
20.6% — 21/102 loci (30 shunts, 391.8 nats; DPANN Nanoarchaeota highways)
Archaeal Informational Core (Ribosomal/RNAP: % w/ HGT)
0.0% — 0/20 loci (0 shunts, 0.0 nats; Fisher's exact p = 0.0235)
Klebsiella 214-kb Core (Lateral HGT % of nats)
0.5% — 77.0 lateral vs. 15,485 vertical nats (99.5% vertical; dgaE +32.9 nats)
0
20
40
60
80
100
120
140
160
Percentage (%)